

Daily Logic Drill #2 — Answers & Investigations

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Implication, Converse, Inverse & Contrapositive · Necessary vs Sufficient.

Section A Rapid Recognition

1. For real numbers x, y , classify “ $x > y$ ” as a condition for “ $x^3 > y^3$ ”.

Answer: Necessary and sufficient.

Method: The function $f(t) = t^3$ has derivative $f'(t) = 3t^2 \geq 0$, with $f'(t) = 0$ only at $t = 0$, so f is strictly increasing everywhere on $\mathbb{R}$. Hence $x > y \iff x^3 > y^3$. Both directions hold.

Investigate further: Unlike squaring ($x > y \not\iff x^2 > y^2$), odd powers preserve order globally across all real numbers.

2. For non-zero real numbers x, y , classify “ $x > y$ ” as a condition for “ $\frac{1}{x} < \frac{1}{y}$ ”.

Answer: Neither necessary nor sufficient.

Method: Not sufficient: $x = 1, y = -2 \implies x > y$, but $\frac{1}{x} = 1 > -\frac{1}{2} = \frac{1}{y}$. Not necessary: $x = -2, y = -1 \implies \frac{1}{x} = -0.5 > -1 = \frac{1}{y}$, while for $x = -1, y = 2$ we have $\frac{1}{x} = -1 < \frac{1}{2} = \frac{1}{y}$ but $x < y$.

Investigate further: The inequality $\frac{1}{x} < \frac{1}{y} \iff \frac{y-x}{xy} < 0$ depends entirely on the sign of the product xy . It only reverses when x and y share the same sign.

3. For the quadratic equation $x^2 - 2kx + 1 = 0$, classify “ $k > 1$ ” as a condition for “the equation has two distinct positive real roots”.

Answer: Necessary and sufficient.

Method: For two distinct real roots, the discriminant $\Delta = 4k^2 - 4 > 0 \iff k^2 > 1 \iff |k| > 1$. By Vieta’s formulas, the product of roots is $1 > 0$, so the roots share the same sign. The sum of roots is $2k$. For the roots to be positive, we need sum $2k > 0 \implies k > 0$. Combining $|k| > 1$ and $k > 0$ gives $k > 1$.

Investigate further: This three-part test (discriminant, product sign, sum sign) is the universal TMUA protocol for quadratic root separation.

4. For real numbers x, y , classify “ $|x + y| = |x| + |y|$ ” as a condition for “ $xy \geq 0$ ”.

Answer: Necessary and sufficient.

Method: Both sides are non-negative, so square both sides: $(x + y)^2 = (|x| + |y|)^2 \iff x^2 + 2xy + y^2 = x^2 + 2|xy| + y^2 \iff xy = |xy|$. Since $xy = |xy| \iff xy \geq 0$, the condition is necessary and sufficient.

Investigate further: The Triangle Inequality $|x + y| \leq |x| + |y|$ achieves equality if and only if x and y have the same sign (or at least one is zero).

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Classify “ $f'(x) > 0$ for all $x \in \mathbb{R}$ ” as a condition for “ f is strictly increasing on $\mathbb{R}$ ”.

Answer: Sufficient but not necessary.

Method: By the Mean Value Theorem, $f'(x) > 0$ everywhere implies $f(b) - f(a) = f'(c)(b - a) > 0$ for all $b > a$, so f is strictly increasing (sufficient). It is not necessary: $f(x) = x^3$ is strictly increasing on $\mathbb{R}$, yet $f'(0) = 0 \not> 0$.

Investigate further: A classic TMUA distinction: strict positivity of the derivative is sufficient, but isolated stationary points do not break strict monotonicity.

6. For real constants a, b, c, d , classify “ $ad - bc \neq 0$ ” as a condition for “the system $ax + by = 0$, $cx + dy = 0$ has only the trivial solution $(x, y) = (0, 0)$ ”.

Answer: Necessary and sufficient.

Method: A homogeneous 2×2 linear system has non-trivial solutions if and only if the lines are collinear, which occurs iff $\det = ad - bc = 0$. Hence having only the trivial solution is equivalent to $ad - bc \neq 0$.

Investigate further: Determinant non-zero $\iff$ unique solution $\iff$ kernel is trivial: the foundation of linear algebra condition strength.

7. For a point $(x, y) \in \mathbb{R}^2$, classify “ $x^2 + y^2 \leq 1$ ” as a condition for “ $|x| + |y| \leq \sqrt{2}$ ”.

Answer: Sufficient but not necessary.

Method: By Cauchy-Schwarz, $(|x| + |y|)^2 \leq (1^2 + 1^2)(x^2 + y^2) = 2(x^2 + y^2)$. If $x^2 + y^2 \leq 1$, then $(|x| + |y|)^2 \leq 2 \implies |x| + |y| \leq \sqrt{2}$ (sufficient). Not necessary: the point $(1, 0.4)$ has $|x| + |y| = 1.4 \leq \sqrt{2} \approx 1.414$, but $x^2 + y^2 = 1 + 0.16 = 1.16 > 1$.

Investigate further: Geometrically, the unit circle is inscribed inside the diamond $|x| + |y| \leq \sqrt{2}$ with four tangent points $(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}})$. The diamond strictly contains the disk.

8. For a positive integer n , classify “ $12 \mid n$ ” as a condition for “ $4 \mid n$ and $6 \mid n$ ”.

Answer: Necessary and sufficient.

Method: An integer n is divisible by both 4 and 6 if and only if n is divisible by $\text{lcm}(4, 6) = 12$. Since $12 \mid n \iff (4 \mid n \wedge 6 \mid n)$, the condition is necessary and sufficient.

Investigate further: Unlike $4 \times 6 = 24$, which is sufficient but not necessary, the LCM gives the exact biconditional.

9. For a triangle with side lengths a, b, c , classify “ $a^2 + b^2 = c^2$ ” as a condition for “the area of the triangle is $\frac{1}{2}ab$ ”.

Answer: Necessary and sufficient.

Method: The area of any triangle with sides a, b and included angle C is $\text{Area} = \frac{1}{2}ab \sin C$. Thus $\text{Area} = \frac{1}{2}ab \iff \sin C = 1 \iff C = 90^\circ$. By the Law of Cosines, $c^2 = a^2 + b^2 - 2ab \cos C$, so $C = 90^\circ \iff a^2 + b^2 = c^2$. Both directions hold.

Investigate further: Pythagoras and the sine area formula link right-angles to area maximization: $\sin C \leq 1$ with equality iff right-angled.

10. For an integer n , classify “ n is odd” as a condition for “ $n^2 \equiv 1 \pmod{8}$ ”.

Answer: Necessary and sufficient.

Method: If n is odd, $n = 2k + 1$, so $n^2 = 4k(k + 1) + 1$. Since $k(k + 1)$ is the product of consecutive integers, it is even, so $4k(k + 1)$ is a multiple of 8, giving $n^2 \equiv 1 \pmod{8}$ (sufficient). Conversely, if n is even, n^2 is even, so $n^2 \not\equiv 1 \pmod{8}$ (necessary).

Investigate further: Every odd square is $1 \pmod{8}$ — one of the most powerful modular invariants in TMUA Paper 2.

Section B Manipulation Drills

1. Determine the necessary and sufficient condition on real k such that the quadratic equation $x^2 - kx + 9 = 0$ has two distinct real roots that are both strictly greater than 1. (after TMUA 2021 Paper 2 Q9)

Answer: $6 < k < 10$.

Method: Let $f(x) = x^2 - kx + 9$. For f to have two distinct real roots > 1 : (1) $\Delta = k^2 - 36 > 0 \implies |k| > 6$. (2) The vertex of the parabola $x = k/2 > 1 \implies k > 2$. Combining with $|k| > 6$ gives $k > 6$. (3) Both roots > 1 requires $f(1) > 0$, so $f(1) = 1 - k + 9 = 10 - k > 0 \implies k < 10$. Together, $6 < k < 10$.

Investigate further: If $k \geq 10$, one root is ≤ 1 ; if $k \leq 6$, the roots are complex or coincident.

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Consider the conditions:

I. $f(-x) = f(x)$ for all $x \in \mathbb{R}$.

II. $f'(-x) = -f'(x)$ for all $x \in \mathbb{R}$.

Classify condition I as a condition for condition II. (after TMUA 2018 Paper 2 Q10)

Answer: Necessary and sufficient.

Method: If I holds, differentiating $f(-x) = f(x)$ via chain rule gives $-f'(-x) = f'(x) \implies f'(-x) = -f'(x)$, so II holds (sufficient). Conversely, if II holds, let $h(x) = f(x) - f(-x)$. Then $h'(x) = f'(x) - (-f'(-x)) = f'(x) + f'(-x) = 0$ for all x , so $h(x) = C$ is constant. At $x = 0$, $h(0) = f(0) - f(0) = 0 \implies C = 0$, so $f(x) = f(-x)$ for all x (necessary).

Investigate further: Because $h(0) = 0$ is guaranteed, the constant of integration vanishes, making evenness and odd derivative strictly equivalent on $\mathbb{R}$.

3. For positive real numbers x, y , classify “ $\ln(x + y) < \ln x + \ln y$ ” as a condition for “ $x > 1$ and $y > 1$ ”.

Answer: Neither necessary nor sufficient.

Method: $\ln(x + y) < \ln(xy) \iff x + y < xy \iff xy - x - y > 0 \iff (x - 1)(y - 1) > 1$. Not sufficient: $x = 1.5, y = 1.5 \implies x > 1$ and $y > 1$, but $(0.5)(0.5) = 0.25 \not> 1$. Not necessary: $x = 10, y = 1.2 \implies (9)(0.2) = 1.8 > 1$, but if $x = 3, y = 0.5$ then $(2)(-0.5) = -1 < 1$. Furthermore, $x > 1$ and $y > 1$ is not necessary because both $x - 1$ and $y - 1$ could never be negative and multiply to > 1 while being positive reals (since $x, y > 0 \implies x - 1 > -1, y - 1 > -1$). So it is neither.

Investigate further: Factoring $xy - x - y = (x - 1)(y - 1) - 1$ converts the logarithmic inequality into a hyperbola region.

4. State the contrapositive of: “If $a^2 + b^2$ is divisible by 3, then both a and b are divisible by 3” (for integers a, b). Is this contrapositive true?

Answer: Contrapositive: “If at least one of a or b is not divisible by 3, then $a^2 + b^2$ is not divisible by 3.” True.

Method: The contrapositive of $P \implies (Q \wedge R)$ is $\neg(Q \wedge R) \implies \neg P \equiv (\neg Q \vee \neg R) \implies \neg P$. In words: if a or b is not divisible by 3, then $a^2 + b^2$ is not divisible by 3. For any integer n , $n^2 \equiv 0$ or $1 \pmod{3}$. Thus $a^2 + b^2 \equiv 0 \pmod{3}$ iff $a^2 \equiv 0$ and $b^2 \equiv 0 \pmod{3} \iff 3 \mid a$ and $3 \mid b$. The original statement is true, so its contrapositive is true.

Investigate further: De Morgan on the conclusion of an implication flips “both” to “at least one fails”.

5. Let T_1 and T_2 be two triangles. Classify “ T_1 and T_2 have equal perimeter and equal area” as a condition for “ T_1 and T_2 are congruent”.

Answer: Necessary but not sufficient.

Method: Congruent triangles have identical sides, so equal perimeter and equal area is necessary. It is not sufficient: a triangle’s area and perimeter fix the semiperimeter s and inradius $r = A/s$, but there exist non-congruent triangles with the same s and r (e.g. non-isometric Heron triangles or right triangles vs oblique triangles sharing s and A).

Investigate further: Two invariants (perimeter and area) are not enough to fix the three degrees of freedom of a triangle up to congruence.

6. For an integer n , classify “ n is odd” as a condition for “ $n^3 - n$ is a multiple of 24”.

Answer: Sufficient but not necessary.

Method: Factor $n^3 - n = (n - 1)n(n + 1)$. If n is odd, $n - 1$ and $n + 1$ are consecutive even integers, so one is divisible by 4 and the other by 2, giving $8 \mid n^3 - n$. Among three consecutive integers, one is always divisible by 3, so $24 \mid n^3 - n$ (sufficient). It is not necessary: if $n = 8$ (even), $8^3 - 8 = 504 = 24 \times 21$, which is divisible by 24.

Investigate further: A classic pitfall: n odd guarantees divisibility by 24, but certain even integers (n a multiple of 24, or $n \equiv 0 \pmod{8}$) also work.

7. Suppose propositions P, Q, R satisfy $P \implies Q$ and $R \implies Q$. Must either $P \implies R$ or $R \implies P$ be true? Justify with a mathematical counterexample.

Answer: No. (Counterexample: let Q be $x^2 \geq 0$, P be $x = 1$, and R be $x = -1$).

Method: Both $x = 1$ and $x = -1$ imply $x^2 \geq 0$, so $P \implies Q$ and $R \implies Q$ are both true. But $x = 1 \implies x = -1$ is false, and $x = -1 \implies x = 1$ is false. Thus neither $P \implies R$ nor $R \implies P$ holds.

Investigate further: Two independent sufficient conditions for the same conclusion need have no logical relation to each other.

8. For real c , determine whether “ $c < 0$ ” is necessary, sufficient, both, or neither for the quartic equation $x^4 + x^2 + c = 0$ to have at least two distinct real roots.

Answer: Necessary and sufficient.

Method: Let $u = x^2 \geq 0$. The equation becomes $u^2 + u + c = 0$. By Vieta's formulas, the product of roots in u is c . If $c < 0$, the roots in u have opposite signs, so there is exactly one positive root $u_1 > 0$, yielding two distinct real roots $x = \pm\sqrt{u_1}$ (sufficient). If $c \geq 0$, then for all $u \geq 0$, $u^2 + u + c \geq c \geq 0$; the only possible real root is $u = 0$ when $c = 0$, which gives $x = 0$ (one root, not two distinct roots). Thus $c < 0$ is also necessary.

Investigate further: Substitution $u = x^2$ requires paying close attention to the domain $u \geq 0$: only positive u generate pairs of real roots.

9. Consider the statement: “If n is a multiple of 9, then n^2 is a multiple of 27” (for positive integer n). State its converse, and determine whether the converse is true or false.

Answer: Converse: “If n^2 is a multiple of 27, then n is a multiple of 9.” True.

Method: The converse states $27 \mid n^2 \implies 9 \mid n$. If $27 \mid n^2$, then $3^3 \mid n^2$. Let the power of 3 dividing n be 3^k . Then 3^{2k} divides n^2 , so $2k \geq 3 \implies k \geq 2$. Thus $3^2 = 9$ divides n . Hence the converse is true.

Investigate further: The exponent 3 in $27 = 3^3$ forces $k \geq \lceil 3/2 \rceil = 2$, which uniquely forces divisibility by $3^2 = 9$.

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable. Classify “ $f'(a) = 0$ and $f''(a) > 0$ ” as a condition for “ f has a local minimum at $x = a$ ”.

Answer: Sufficient but not necessary.

Method: By the Second Derivative Test, $f'(a) = 0$ and $f''(a) > 0$ guarantees a local minimum (sufficient). It is not necessary: $f(x) = x^4$ has a global minimum at $x = 0$, where $f'(0) = 0$ and $f''(0) = 0 \not> 0$.

Investigate further: Higher-order flat minima (x^4, x^6) refute necessity of strict second derivatives.

Section C Substitution & Structure

1. For real numbers x, y , the condition “ $x + y$ is an integer and $x - y$ is an integer” is:

Answer: A) necessary but not sufficient for “ x and y are both integers”.

Method: Necessary: if x, y are both integers, then $x + y$ and $x - y$ are automatically both integers. Not sufficient: $x = y = 0.5$ gives $x + y = 1$ and $x - y = 0$, both integers, but x, y themselves are not — ruling out options B and C. Necessity alone rules out D.

Investigate further: Solving $x = \frac{(x+y)+(x-y)}{2}$, $y = \frac{(x+y)-(x-y)}{2}$ shows exactly what goes wrong: x, y are only forced to be integers when $x + y$ and $x - y$ have the *same parity* (so the sum/difference is even) — a condition the question's hypothesis never guarantees.

2. Determine the condition on real parameters p, q that is necessary and sufficient for the cubic $x^3 - 3px + q = 0$ with $p > 0$ to have three distinct real roots.

Answer: B) $4p^3 > q^2$.

Method: Let $f(x) = x^3 - 3px + q$. Differentiating gives $f'(x) = 3x^2 - 3p = 3(x^2 - p) = 0 \implies x = \pm\sqrt{p}$. A cubic with positive leading coefficient has three distinct real roots iff its local maximum is positive and local minimum is negative: $f(-\sqrt{p}) > 0$ and $f(\sqrt{p}) < 0$, which is equivalent to $f(-\sqrt{p})f(\sqrt{p}) < 0$. We

compute: $f(\sqrt{p}) = p^{3/2} - 3p^{3/2} + q = q - 2p^{3/2}$, and $f(-\sqrt{p}) = -p^{3/2} + 3p^{3/2} + q = q + 2p^{3/2}$. Their product is $(q - 2p^{3/2})(q + 2p^{3/2}) = q^2 - 4p^3 < 0 \iff 4p^3 > q^2$. This matches option B.

Investigate further: This is the classic cubic discriminant $\Delta = 4p^3 - q^2 > 0$ derived entirely from calculus extrema.

3. Classify “a triangle has all three altitudes of equal length” as a condition for “the triangle is equilateral”. Justify using the fact that an altitude to side a has length $\frac{2 \times \text{Area}}{a}$.

Answer: C) necessary and sufficient.

Method: Let the altitudes to sides a, b, c be h_a, h_b, h_c , and let the triangle’s area be A . Sufficient: if the triangle is equilateral, $a = b = c$, so $h_a = \frac{2A}{a} = \frac{2A}{b} = \frac{2A}{c} = h_b = h_c$ directly. Necessary: if $h_a = h_b = h_c$, then $\frac{2A}{a} = \frac{2A}{b} = \frac{2A}{c}$, and since $A \neq 0$, this forces $a = b = c$ — the triangle is equilateral.

Investigate further: Both directions rely on the same identity ($h_a = \frac{2A}{a}$) read in opposite directions — side-length equality and altitude-length equality are simply the same fact viewed through a common factor of $2A$.

4. For positive integers a, b, n , suppose you’re told only that n divides a , or n divides b (possibly both). Does this pin down whether n divides the product ab ? And working the other way: does knowing n divides ab pin down whether n divides a or n divides b ? (after TMUA 2019 Paper 2 Q5)

Answer: B) “ n divides a or n divides b ” is sufficient but not necessary for “ n divides ab ”.

Method: Sufficient: if $n \mid a$ (or $n \mid b$), then n divides the product ab directly. Not necessary: taking $n = 9, a = 3, b = 3$ gives $ab = 9$, divisible by 9, but neither a nor b individually is divisible by 9 — ruling out options A and C. Option D is ruled out by the sufficiency direction alone.

Investigate further: This only fails because $n = 9 = 3^2$ is not squarefree — if n were prime, divisibility of a product would force divisibility of a factor (Euclid’s Lemma), making the condition necessary too. The gap here is exactly why primality matters so much in Number Theory.

5. For a twice-differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, which of the following is necessary and sufficient for the midpoint convexity condition $\frac{f(a)+f(b)}{2} \geq f\left(\frac{a+b}{2}\right)$ to hold for all $a, b \in \mathbb{R}$? (after TMUA 2018 Paper 2 Q18)

Answer: B) $f''(x) \geq 0$ for all $x \in \mathbb{R}$.

Method: For a twice-differentiable function, convexity is equivalent to non-negative second derivative: $f''(x) \geq 0$ for all x . Option A ($f''(x) > 0$) is sufficient but not necessary: $f(x) = x^4$ is convex everywhere but $f''(0) = 0$, and linear functions $f(x) = mx + c$ satisfy equality with $f''(x) \equiv 0$. Option C and D have no relation to convexity. Hence B is necessary and sufficient.

Investigate further: Midpoint convexity plus continuity implies full Jensen convexity $\sum \lambda_i f(x_i) \geq f(\sum \lambda_i x_i)$.

6. Classify “ n^2 is even” as a condition for “ n is even”.

Answer: C) necessary and sufficient.

Method: Sufficient: if n^2 is even, then n cannot be odd (an odd number squared is odd), so n is even. Necessary: if n is even, n^2 is even directly.

Investigate further: Both directions holding here is exactly why “ n even $\iff n^2$ even” is such a reliable tool in proofs by contradiction.

7. Let $p(x)$ be a cubic polynomial with real coefficients. Which of the following conditions is necessary and sufficient for the existence of $c \in (a, b)$ such that $p'(c) = 0$? (after TMUA 2021 Paper 2 Q8)

Answer: C) $p'(x) = 0$ has real roots and at least one lies in (a, b) .

Method: The statement asserts the existence of a real number $c \in (a, b)$ satisfying $p'(c) = 0$. By definition, this occurs if and only if $p'(x) = 0$ has real roots and at least one root lies in the open interval (a, b) . Option A ($p(a) = p(b)$) is sufficient by Rolle’s Theorem, but not necessary (a cubic can have a stationary point in (a, b) without $p(a) = p(b)$). Option B is sufficient by the Intermediate Value Theorem, but not necessary (both $p'(a)$ and $p'(b)$ could be positive while a local min and max both lie inside (a, b)). Thus C is the exact necessary and sufficient condition.

Investigate further: Distinguishing between a theorem’s hypothesis (sufficient, like Rolle’s) and the exact semantic equivalence is a core TMUA Paper 2 theme.

8. State the contrapositive of “If a and b are both irrational, then ab is irrational.” Is the original statement true? Give a counterexample if false.

Answer: Contrapositive: “If ab is rational, then a is rational or b is rational.” The original statement is false; counterexample $a = b = \sqrt{2}$.

Method: $\sqrt{2}$ is irrational, but $ab = \sqrt{2} \times \sqrt{2} = 2$, which is rational — disproving the original. Since the contrapositive always shares the original’s truth value, the same counterexample shows the contrapositive is false too: $ab = 2$ is rational, yet neither $a = \sqrt{2}$ nor $b = \sqrt{2}$ is rational.

Investigate further: One counterexample disproves both a statement and its contrapositive simultaneously — there is never a need to separately hunt for a second one.

Section D Challenge

1. Let $f(x) = x^2 + bx + c$ with $f(0) = c$. Let P be the statement: “if $c < 0$, then f has two distinct real roots.” Which of the following must be true? (after TMUA 2022 Paper 2 Q5)

Answer: F) I and III only.

Method: I (P): the discriminant is $b^2 - 4c$; if $c < 0$ then $-4c > 0$, so $b^2 - 4c > b^2 \geq 0$, meaning the discriminant is always strictly positive — f always has two distinct real roots. P is true. II (converse): “if f has two distinct real roots, then $c < 0$ ” — false, since $b = 3, c = 1$ gives discriminant $9 - 4 = 5 > 0$ (two distinct real roots) with $c = 1 \not< 0$. III (contrapositive): shares P ’s truth value, so it is true.

Investigate further: A true original implication with a false converse and a true contrapositive is the quintessential TMUA implication pattern.

2. Consider a finite list of n real numbers (repeats allowed), sorted into non-decreasing order. Let P be the statement “ n is odd”, and let Q be the statement “the middle value of the sorted list equals one of the list’s own entries”. P is: (after TMUA 2022 Paper 2 Q6)

Answer: B) sufficient but not necessary for Q .

Method: Sufficient: if n is odd, the list has a unique middle term when sorted, and that middle term is

the median — a list member. Not necessary: the list 4, 4 has $n = 2$ (even), and the median is $\frac{4+4}{2} = 4$, which is a list member — so Q holds even though P fails, ruling out options A and C. Sufficiency alone rules out D.

Investigate further: Any even-length list whose two middle entries are equal gives the same kind of counterexample — it is n being odd that *guarantees* Q , not the only way to achieve it.

3. For a positive integer n , let P be “ n leaves remainder 1 when divided by 4” and Q be “ n^2 leaves remainder 1 when divided by 8.” P is:

Answer: B) sufficient but not necessary for Q .

Method: Sufficient: if $n = 4k + 1$, then $n^2 = 16k^2 + 8k + 1$, and modulo 8 this is 1 (both $16k^2$ and $8k$ are multiples of 8). Not necessary: if $n = 4k + 3$, then $n^2 = 16k^2 + 24k + 9 = 16k^2 + 24k + 8 + 1$, and modulo 8 this is also 1 (all of $16k^2, 24k, 8$ are multiples of 8) — so Q can hold with $n \equiv 3 \pmod{4}$, where P fails, ruling out A and C. Sufficiency alone rules out D.

Investigate further: The true necessary-and-sufficient condition for Q is simply “ n is odd” — every odd square is $\equiv 1 \pmod{8}$, regardless of which odd residue mod 4 it comes from.

4. Statement S : “If a quadrilateral has four equal sides, then its diagonals are perpendicular.” (a) Is S true? (b) State S ’s converse. Is the converse true?

Answer: (a) True (a rhombus). (b) Converse: “If a quadrilateral’s diagonals are perpendicular, then it has four equal sides.” The converse is false.

Method: (a) Let quadrilateral $WXYZ$ have $WX = XY = YZ = ZW$, and let M be the intersection of diagonals WY and XZ . Triangles WXY and WZY are congruent (SSS: $WX = WZ$, $XY = ZY$, WY shared), so $\angle XWY = \angle ZWY$, meaning WY bisects $\angle XWZ$; since triangle WXZ is isosceles ($WX = WZ$), this bisector is also the perpendicular bisector of XZ . So the diagonals meet at right angles. (b) A kite with $WX = WZ$ but $XY = ZY \neq WX$ (two pairs of adjacent equal sides, not all four equal) has the same SSS congruence argument still apply to triangles WXY, WZY — so its diagonals are still perpendicular, but it is a counterexample to the converse since its sides are not all equal.

Investigate further: “Four equal sides” is a strictly stronger condition than “diagonals perpendicular” — every rhombus is a kite, but not every kite is a rhombus.

5. Statement S : “If n is a positive integer such that $n^2 - 1$ is prime, then $n = 2$.” (a) Is S true? Prove or disprove. (b) State S ’s contrapositive. (c) Is “ $n^2 - 1$ is prime” necessary and sufficient for “ $n = 2$ ”?

Answer: (a) True. (b) “If $n \neq 2$, then $n^2 - 1$ is not prime.” (c) Yes.

Method: (a) Factor $n^2 - 1 = (n - 1)(n + 1)$. For a product of two positive integers to be prime, one factor must equal 1. Since $n \geq 1$, the smaller factor is $n - 1$, forcing $n - 1 = 1$, i.e. $n = 2$; checking, $n = 2$ gives $n^2 - 1 = 3$, which is indeed prime. For every other $n > 2$, both $n - 1 \geq 2$ and $n + 1 \geq 4$, so $(n - 1)(n + 1)$ is composite. So $n = 2$ is true, and it is the unique positive integer. (b) Negate both clauses of S and swap. (c) Since $n = 2$ is the unique solution, “ $n^2 - 1$ prime” and “ $n = 2$ ” describe exactly the same set — necessary and sufficient.

Investigate further: Uniqueness turns a one-directional conditional implication into a biconditional equivalence for free.

Top 5 Patterns Today

1. **Directional Testing of Necessity vs Sufficiency.** “ P is sufficient for Q ” means $P \implies Q$; “ P is necessary for Q ” means $Q \implies P$. Always evaluate the forward and backward implications independently. (see A1, A3, A5, B1, B5)
2. **Strict Inclusions and Counterexample Witnesses.** A condition that is strictly more restrictive is sufficient but not necessary (e.g. $f'(x) > 0$ for strict monotonicity, unit disk in $|x| + |y| \leq \sqrt{2}$). Disprove necessity with an edge counterexample. (see A5, A7, B6, B10, D2)
3. **Equivalence in Geometry and Algebra.** $P \iff Q$ holds when both directions are valid: altitude equality forces equilateral sides, and symmetric quadratic discriminants pin down positive roots. (see A3, A4, A6, A9, C3)
4. **Polynomial and Cubic Root Separation.** Distinct roots in intervals or cubic extrema requiring $4p^3 > q^2$ demand checking discriminant signs, axis vertices, and interval boundaries together. (see A3, B1, B8, C2, D1)
5. **Divisibility and Modular Invariants.** $n^2 \equiv 1 \pmod{8} \iff n$ odd; $n^3 - n \equiv 0 \pmod{24}$ for odd n . Exponent parity in prime factors establishes exact divisibility equivalences. (see A8, A10, B4, B9, D3)

Common Traps to Avoid

1. **Assuming Sufficiency Automatically Implying Necessity.** $f'(x) > 0$ guarantees f is strictly increasing, but $f(x) = x^3$ is strictly increasing with $f'(0) = 0$; never assume the converse holds without checking. (see A5, B6, B10, D2, D3)
2. **Overlooking Sign Flips in Reciprocals and Inequalities.** $x > y$ does not imply $1/x < 1/y$ when signs differ ($x = 1, y = -2$); sign preservation is essential when inverting inequalities. (see A2, B3, C1, D4)
3. **Assuming False Transitivity Across Shared Consequences.** $P \implies Q$ and $R \implies Q$ does not imply any logical link between P and R ($x = 1$ and $x = -1$ both square to 1). (see B7, C4)
4. **Confusing Contrapositive with Converse.** The contrapositive of $a^2 + b^2 \equiv 0 \pmod{3}$ always shares its truth value; the converse requires its own independent justification. (see B4, B9, C8, D1, D4)
5. **Missing Multiplicity or Trivial Degeneracies.** In $x^4 + x^2 + c = 0$, $c = 0$ yields $x = 0$ with multiplicity 2 rather than distinct real roots; strict inequalities at boundaries make or break equivalence. (see A3, B1, B8, C7, D5)

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths